

# Business Requirement Document (BRD)

ContractAI Platform (MVP – 1 Month Build)\*\*

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# 1. Executive Summary

ContractAI is an AI-powered contract intelligence and automation platform that enables enterprises to **ingest, classify, analyze, extract, compare, and monitor contractual information** using advanced NLP/LLM models. The platform will serve Legal, Procurement, Compliance, and Business teams by reducing contract review time, increasing compliance, and mitigating risks.

This BRD outlines the **functional, non-functional, technical, and operational requirements** for the ContractAI MVP to be delivered in **1 month** by:

- **2 AI Engineers / Full-Stack Engineers**
- **1 Software Architect**

The MVP focuses on **core contract ingestion → extraction → approval workflow → dashboard** capabilities.

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## 2. Project Objectives

### Primary Objectives

1. Automate extraction of key clauses & metadata from contracts (PDF, Word).
2. Compare contracts against templates or past versions.
3. Provide AI-driven risk scoring and deviation detection.
4. Provide a workflow for legal review and approval.
5. Build an admin dashboard for tracking contract statuses.

### Secondary Objectives

1. Enable conversational contract querying (Chat with Contract).

2. Generate summaries & obligation checklists.
  3. Integrate with email/upload sources.
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## 3. Scope Definition

### In-Scope (MVP)

- User authentication & role-based access (Legal, Procurement, Admin).
- Contract upload (PDF, DOCX).
- OCR & text extraction pipeline.
- AI-driven clause extraction (20–25 clause types).
- Contract classification (NDA, MSA, PO, SOW, Lease, Employment).
- Template deviation detection.
- Risk scoring model.
- Contract summary generation (LLM).
- Standard approval workflow (Upload → AI Review → Human Review → Final Approval).
- Dashboard & metrics.
- Audit logs and contract repository.

### Out-of-Scope (Not part of 1-month MVP)

- E-signature workflows (DocuSign/AdobeSign integration).
- Multi-language contract support.

- Complex ERP/CRM integrations (SAP/Oracle/Salesforce).
  - Multi-tenant SaaS billing layer.
  - Continuous compliance monitoring (phase 2).
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## 4. Stakeholder Analysis

| Stakeholder         | Role                  | Needs                                         |
|---------------------|-----------------------|-----------------------------------------------|
| Legal Team          | Primary user          | Faster contract review, clear risk indicators |
| Procurement         | Frequent uploader     | Clause compliance to supplier guidelines      |
| Business/Operations | Secondary             | Faster cycle time                             |
| Compliance / Audit  | Reviewer              | Deviation logs, audit trails                  |
| CTO/Architect       | Technical owner       | Scalable + secure                             |
| AI Engineers        | Implement AI features | Clear models + data flows                     |

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## 5. Functional Requirements

### 5.1 User Management

- FR-1: Users can register/login via email/password.
  - FR-2: Admin can create roles: Admin, Legal Reviewer, Proc Reviewer, Viewer.
  - FR-3: Role-based access restrictions.
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## 5.2 Contract Upload & Ingestion

- FR-4: Users can upload files (PDF, DOCX, PNG/JPG images).
  - FR-5: Max upload size: 25MB.
  - FR-6: OCR must run on scanned PDFs using Tesseract/DocTR.
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## 5.3 Contract Classification

- FR-7: AI auto-classifies contracts into predefined types (NDA, MSA, SOW, PO, etc.).
  - FR-8: Confidence score displayed.
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## 5.4 Clause Extraction

- FR-9: AI extracts 20–25 standard clauses, e.g.:
  - Term & Termination
  - Confidentiality
  - Liability
  - Indemnity
  - Governing Law
  - Payment Terms
  - Force Majeure
  - Non-Compete
- FR-10: Output must highlight extracted text spans.

- FR-11: Extraction accuracy  $\geq 85\%$  on curated sample set.
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## 5.5 Deviation & Risk Analysis

- FR-12: Compare extracted clauses against **standard template clauses**.
  - FR-13: Highlight deviations in red/yellow/green.
  - FR-14: Generate a Risk Score (Low/Medium/High).
  - FR-15: Provide explanation of risk reasons (LLM-based).
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## 5.6 AI-Generated Summary

- FR-16: Auto-generate executive summary.
  - FR-17: Auto-generate obligation checklist.
  - FR-18: Allow user to regenerate summary with custom prompts.
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## 5.7 Contract Repository

- FR-19: All contracts stored with version history.
  - FR-20: Repository searchable by:
    - Contract name, vendor, date, clause keywords, risk level, status.
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## 5.8 Contract Review Workflow

- FR-21: Stages → Uploaded → AI Reviewed → Legal Reviewed → Approved.
  - FR-22: Reviewer can mark clauses as correct/incorrect.
  - FR-23: Comments + tagging + recommendations.
  - FR-24: Final approval note captured.
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## 5.9 Conversational Search (Smart Query)

- FR-25: Chat-like interface to ask questions:
    - “What is the liability cap?”
    - “Does this contract include auto-renewal?”
  - FR-26: LLM must cite extracted clause references.
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## 5.10 Dashboard & Reporting

- FR-27: Dashboard KPIs:
    - No. of contracts processed
    - Approval cycle time
    - High-risk contracts
    - Contract type distribution
  - FR-28: Downloadable reports (CSV, PDF).
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# 6. Non-Functional Requirements

## 6.1 Performance

- NFR-1: Contract upload → AI analysis ≤ **20 seconds** (for 10-page doc).
- NFR-2: Dashboard load time ≤ **2 seconds**.

## 6.2 Security

- NFR-3: JWT-based authentication.
- NFR-4: Data encrypted at rest (AES256) & in transit (TLS 1.2+).
- NFR-5: Role-based access enforcement.

## 6.3 Scalability

- NFR-6: Support 5,000 contracts/month in MVP.
- NFR-7: Stateless backend with containerization.

## 6.4 Reliability

- NFR-8: System uptime ≥ 99%.
- NFR-9: Auto-restart capabilities in failure.

## 6.5 Compliance

- NFR-10: GDPR/ISO27001-ready architecture.
- NFR-11: Full audit logging.

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# 7. Architecture Overview (High-Level)



## 7.1 Proposed Stack

### Frontend

- React.js / Next.js
- Tailwind / Material UI
- Axios, Redux Toolkit (optional)

### Backend

- FastAPI / Node.js (Express)
- Microservices for:
  - Document processing
  - LLM extraction
  - Workflow service
  - Authentication

### AI Components

- Open-source LLM (Llama 3 / Mistral)
- RAG pipeline with vector DB
- OCR: Tesseract / DocTR
- Clause extraction models: fine-tuned BERT/DeBERTa

### Database

- PostgreSQL (contracts + users)
- Milvus / Pinecone (vector embeddings)

## Deployment

- Docker + Kubernetes (single-node cluster possible)
  - CI/CD with GitHub Actions
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## 8. Data Flow Diagram

### 1. Upload →

PDF/DOCX → OCR → Text cleaned → Embeddings → Clause extraction → Classification

### 2. AI Layer →

Compare with template → Risk scoring → Summary generation

### 3. UX Layer →

Dashboard → Review workflow → Approval → Repository

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## 9. Development Plan (1-Month Timeline)

### Week 1

- Set architecture, repo, CI/CD
- Build ingestion pipeline (OCR, parsing)
- LLM integration
- Database schema design

### Week 2

- Clause extraction models
- Template comparison engine

- Contract classification
- Frontend screens (upload, viewer)

### **Week 3**

- Risk scoring engine
- Summary + obligation generator
- Workflow implementation
- Dashboard with KPIs

### **Week 4**

- Integrations & UX polish
- Performance optimization
- Security hardening
- QA, UAT, Deployment

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## **10. UI/UX Requirements**

### **Required Screens**

1. **Login/Signup**
2. **Contract Upload Page**
3. **Contract Viewer with Clause Highlights**
4. **AI Insights Panel (Summary, Risk, Deviations)**
5. **Review Workflow Page**

- 6. **Dashboard**
- 7. **Repository Search**
- 8. **Admin Settings**

Design Style: **Sleek, modern enterprise, minimalistic**, similar to:

- Ironclad
- Kira Systems
- DocuSign Insight

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## 11. Acceptance Criteria

| Requirement             | Acceptance Criteria                          |
|-------------------------|----------------------------------------------|
| Clause Extraction       | ≥85% accuracy on 50-sample dataset           |
| Upload-to-Insights Time | ≤20 seconds                                  |
| Risk Score              | Correctly reflects deviations                |
| Summary                 | Covers purpose, term, key risks, obligations |
| Dashboard               | Shows correct metrics                        |
| Workflow                | Reviewers can approve/reject, add comments   |
| Security                | Access controlled by roles                   |

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## 12. Risks & Mitigations

| Risk       | Impact | Mitigation              |
|------------|--------|-------------------------|
| OCR errors | Medium | Use DocTR, clean images |

|                                |        |                                       |
|--------------------------------|--------|---------------------------------------|
| Low clause extraction accuracy | High   | Use curated training set              |
| LLM hallucinations             | Medium | Force extraction → evidence grounding |
| Time crunch                    | High   | Strict scope control                  |

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## 13. Future Enhancements (Post-MVP)

1. Multi-language contract support
2. E-signature integration
3. Auto-obligation tracking & reminders
4. SAP/Salesforce integration
5. Multi-tenant SaaS architecture
6. Custom clause training for each organization